SIMATS ENGINEERING

ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Total Marks: 25

Code of Conduct

I C. Mokshith (192525165) (Name / Reg. No.) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: C. Mokshith

Assessment Tasks
Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of **data centers** in delivering cloud services..
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Q) Define cloud computing and explain any four characteristics that make it diffent from traditional and premise computing.

A) Cloud computing is the deliocry of computing resources such as serves, storage, networking database, and software over the internet Instead of purchasing and maintaining Physical in frestmetuns, User can access resources from cloud provides whenever required.

Four characteristis!-

1. on - dem and self services : User can create or access computing resources whenever required without direct into raction with the service provider.

2.) Broad network access!-

cloud services can be accessed through the internet using devices such as computers, tablets, and smart phones.

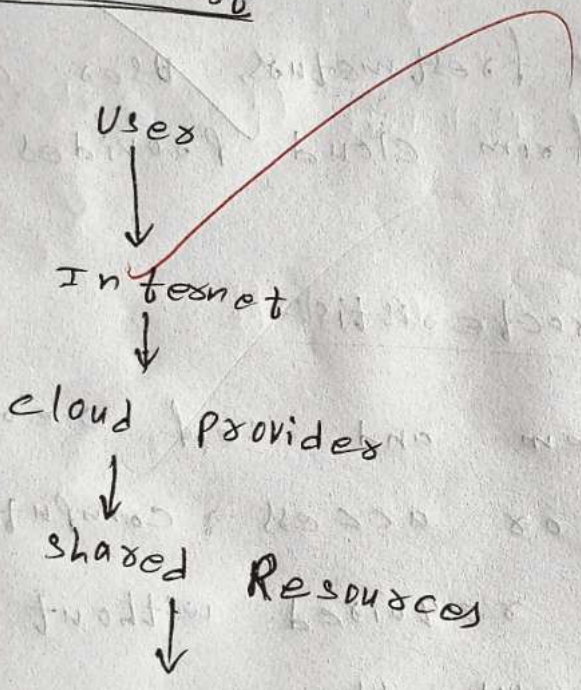
3.) Resource Pooling:-

The provider shares a common pool of computing resource among multiple users using virtualization.

4) Rapid elasticity:-

Resources can be increased or decreased quickly according to demand.

Flow chart



1) Trace the evaluation of cloud computing from hard. wave evoluation internet software evolvation, and show how these changes enabled modern cloud platform.

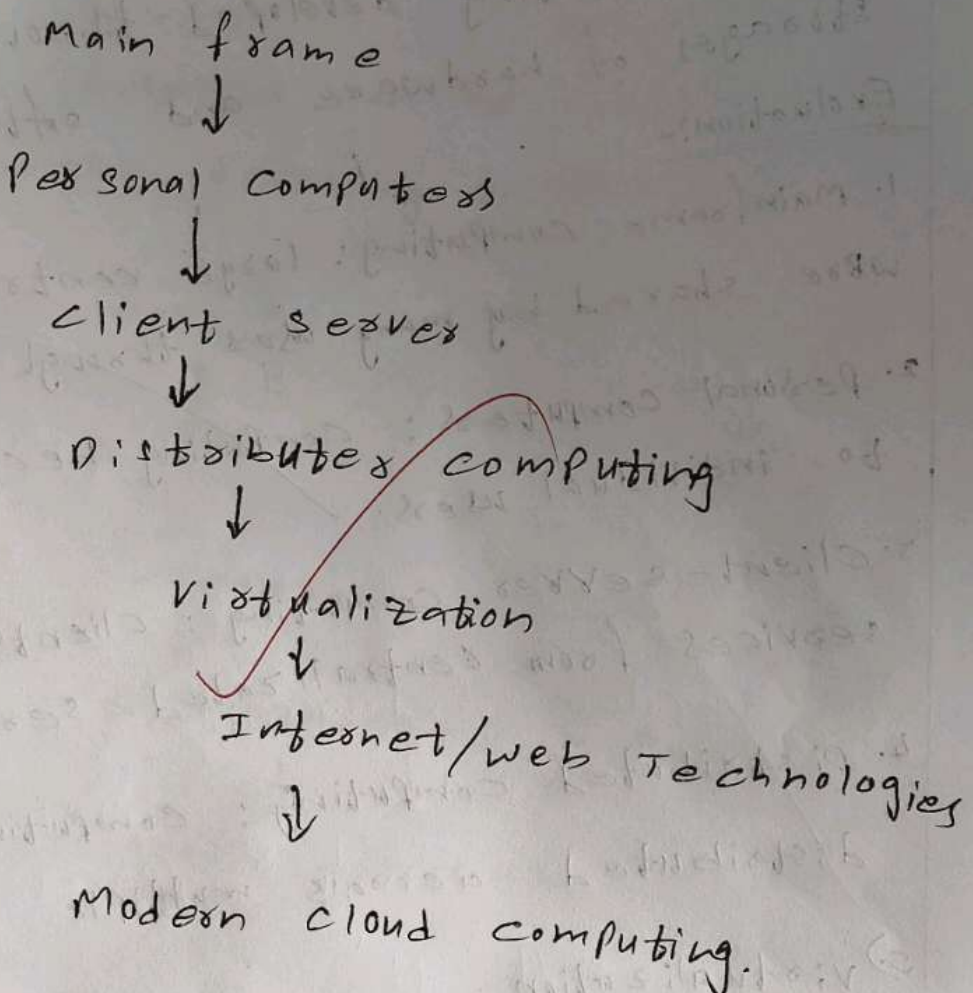
A) Cloud computing developed through several storages of hardware and software evoluation.

Evoluation:-

1. Mainframe computing: large centralization computes were shared by many uses through terminals.
2. Pesonal computers: computing becouse available to individual uses.
3. Client-server computing: clients requesting services from centralizated servers.
4. Distributed computing: computing tasks were distributed acoross multiple connected computers
5. virtualization: one physical server could run multiple virtual machines, improving resource utilization virtualization.
6. Internet and web evoluation:- High speed in internet and web tech role gies made remote access to com puting resources possible.

7) Cloud computing:- providers began offering computing resources as scable services over than Internet.

Flow chart



3.) Explain the role of server virtualization in cloud computing Differentiable between virtulization as an enabling technology and cloud computing as aservice model.

A.) server virtualization is a technology that divides one physical server into multiple idependent virtual machines server into multiple (VMs) wing a software layer called hypervisor.

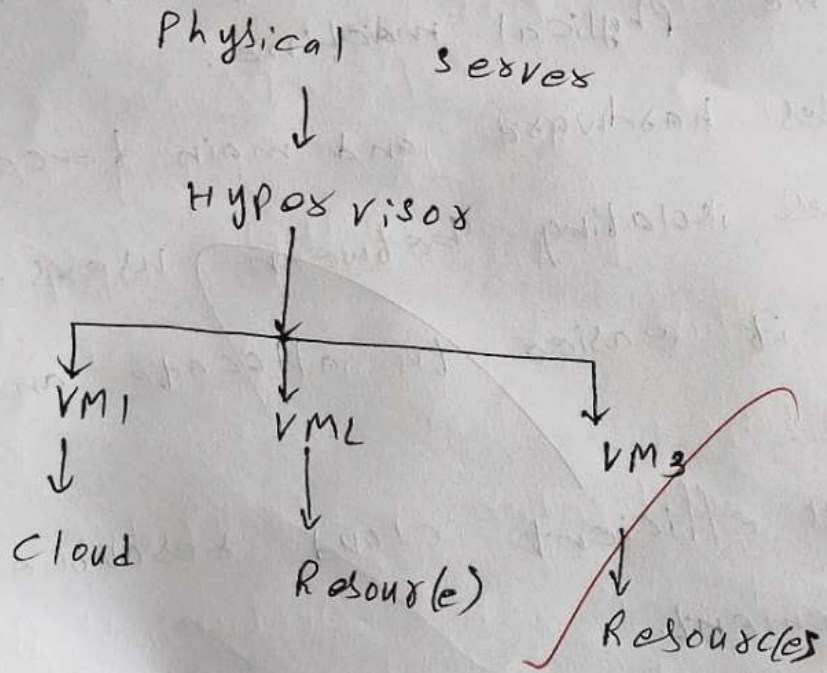
Role in cloud competing:-

→ Im proves utilization of physical sorres
→ Allows multiple operating system to run on one physical machine.
→ Reduces hardware and main tence costs.
→ Provides isolating between users.
→ Makes it easies to allocate and scale resouslos.
→ supports efficient cloud resource management.

Virtulization vs cloud computing:-

Virtulization	cloud computing.
→ It is a enabling technology	→ It is a services delivery modes.
→ creates resources from Physical resources	→ Delivers computing resources through a network.
→ mainly focuses on resources abstraction and utilization	→ focuses on Providing resources as services.

Flowchart



Discuss the role of data centers in delivering services.

A) A data center is a facility containing computing infrastructure such as servers, storage system, networking equipment, power system, and cooling system.

Major roles:-

1.) Computing: Provides physical servers to run application and cloud services.

2) Storage:- Stores user data, applications, databases, and backs.

3.) Networking:- connects servers and users through high speed networks.

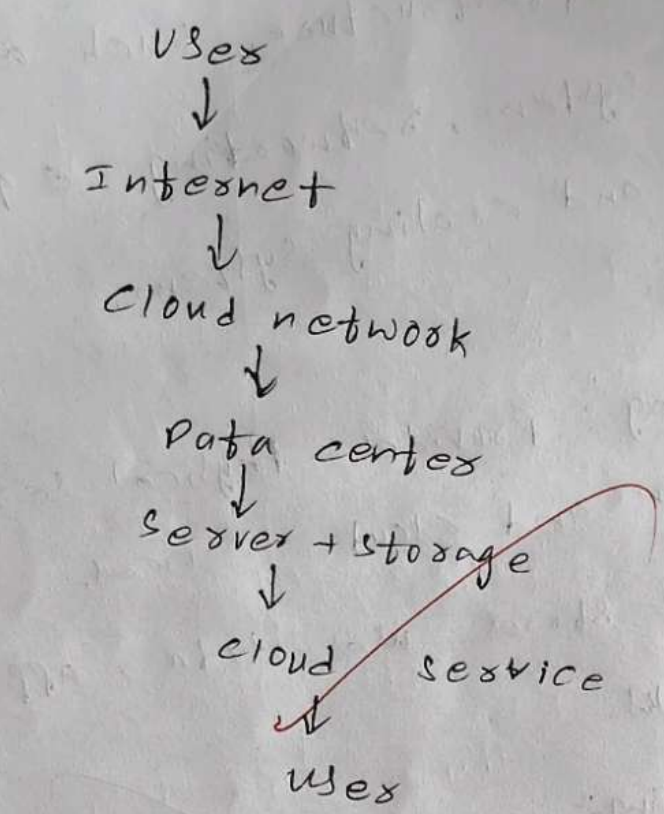
4.) Virtualization:- Supports virtual machine and efficient resource sharing.

5.) Security:- Uses physical and network security mechanisms to protect resources.

6.) Availability:- Uses backup power, redundant system, and multiples server to maintain service service availability.

7.) Scalability:- Allows cloud provides to add more hardware when demand increase.

Flow Chart



5) Different Iaas, Paas, and Xcee's with one suitable real-world example for each model.

A) cloud services can be provided at different levels depending on how much the cloud provider manages.

model	Full form	what is provides
Iaas	Infrastructure as a service	virtual machines, storage, networking
Paas	Platform aa a service	Platform, runtime, development tools
Saas	softwares aa a service	Ready-to use softwa
Xaas	Anything/Everything aa a services	various resources or services delivered through the cloud.

1.) Iaas :-

Provides basic computing infrastru such as virtual servers, storage, and networking The user manages the operati system and applications.

2.) Paas :-

Provides a ready-made platform from developing and deploying applications Developers can focus mainly on applications development instead of managing infrastructure

3) **SaaS:-** Provides complete software applications through the internet. Users generally only need to use the applications.

4) **XaaS:-** A broad term for delivering different type of IT capbilities as services over the internet.

10/10/26